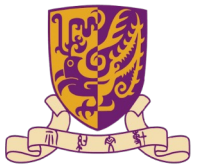
香港中文大學
The Chinese University of Hong Kong

合见工软产学融合赛题
Thermal-aware 3D Floorplanning Placer
An Optimization Solution Based on FM-ILP Hybrid Algorithm

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Intelligent DEsign Automation Lab (IDEALab)
Supervisor: Prof. Tsung-Yi Ho
The Chinese University of Hong Kong
2025-10-11

Outline

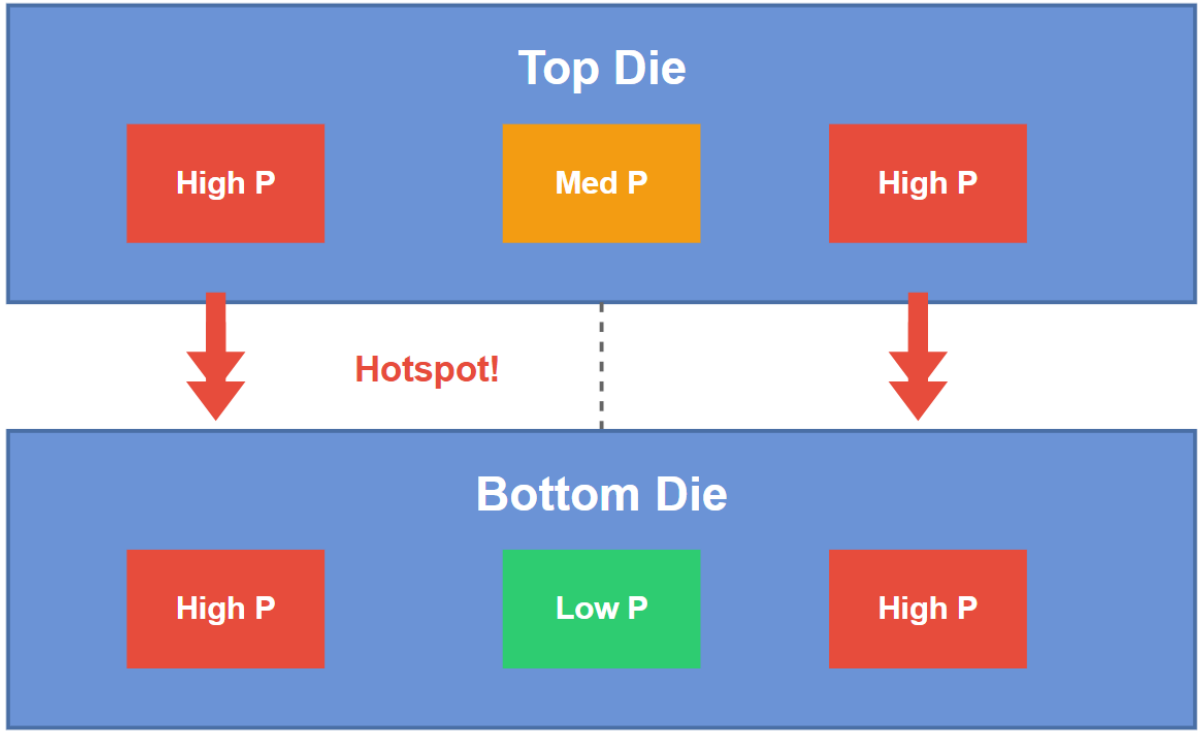


- Background
- Formulation
- Algorithm Design
- Experiments

Thermal Challenges in 3D ICs

- 3D stacking extends Moore‘s Law;
- Vertical integration improves performance and density;
- TSV or hybrid bump interconnects enable inter-layer communication

- Vertical heat conduction;
- Cross-layer thermal coupling;
- Thermal impact – warpage etc.



Formulation

Input:
- Blocks Netlist
- Power

Output:
- Die partition
- 3D layout

Objective:
- Co-optimize the thermal and wirelength

Objective Function

minimize: $\alpha \cdot \text{HPWL} + \beta \cdot \text{Thermal_Penalty}$

$\text{HPWL} = \Sigma\, \text{weight}_{ij} \cdot (|x_i - x_j| + |y_i - y_j|)$

Decision Variables

- $x_i, y_i \in [0, W] \times [0, H]$ (module coordinates)
- $r_i \in \{0, 1\}$ (rotation flag)

Constraints

√ Boundary: $x_i \geq 0.1\text{mm}$

√ Non-overlap (same layer)

√ Min spacing: 100μm

√ Thermal: z_{ij} constraints

Thermal Penalty Components

Intra-layer

- Same die, high power
- Distance < threshold
- Penalty $\propto (P_i + P_j)$

Cross-layer

- Top ↔ Bottom
- $z_{ij} = 1$ if overlap
- Penalty $= (P_i + P_j) \cdot z_{ij}$

Algorithm Design

Objective: $\min \alpha \cdot \sum_{(i,j)\in\mathcal{E}} C_{ij} \cdot (|c^x_i - c^x_j| + |c^y_i - c^y_j|) + \beta \cdot \Phi_{thermal} + \gamma \cdot \Delta P$

Decision Variables

Position:

$x_i, y_i \in [0, W] \times [0, H]$ (continuous)

Die Assignment:

$z_i \in \{0, 1\}$ (0=bottom, 1=top)

Rotation:

$r_i \in \{0, 1\}$ (0=H, 1=V, for non-square)

Key Definitions

Module Center:

$c_i = (x_i + w^{eff}_i/2, y_i + h^{eff}_i/2)$

Effective Dimensions:

$w^{eff}_i = w_i \cdot (1 - r_i) + h_i \cdot r_i$

$h^{eff}_i = h_i \cdot (1 - r_i) + w_i \cdot r_i$

Connection Strength:

C_{ij}: wirelength weight between *i* and *j*

Penalty Terms

- $\Phi_{thermal}$: Thermal penalty accounting for:
  - Intra-layer hotspots (high-power modules too close on same die)
  - Cross-layer thermal coupling (overlapping high-power modules)
- ΔP: Inter-layer power imbalance
  - Balances power distribution between top and bottom dies

Algorithm Design

Algorithm 1 ILP-Greedy Hybrid Solver

Require: Thermal blocks $\mathcal{B}$, region size (W, H), simple nets $\mathcal{N}_s$, detailed nets

Ensure: Block locations, orientations, die assignment

$T_1 \leftarrow 0.95 \times T$, $T_2 \leftarrow 0.05 \times T$

// **Phase 1: Global ILP Optimization**

$(\mathcal{L}, \mathcal{O}_c, \mathcal{D}) \leftarrow$ PHASE1-ILP$(\mathcal{B}, \mathcal{N}_s, W, H, T_1)$

// $\mathcal{L}$: locations, $\mathcal{O}_c$: coarse orientations (H/V), $\mathcal{D}$: die assignment

if detailed nets available **and** $T_2 > 0$ **then**

  // **Phase 2: Pin-Aware Greedy Refinement**

  $\mathcal{O}_f \leftarrow$ PHASE2-GREEDY$(\mathcal{B}, \mathcal{L}, \mathcal{O}_c, \mathcal{N}_d, T_2)$

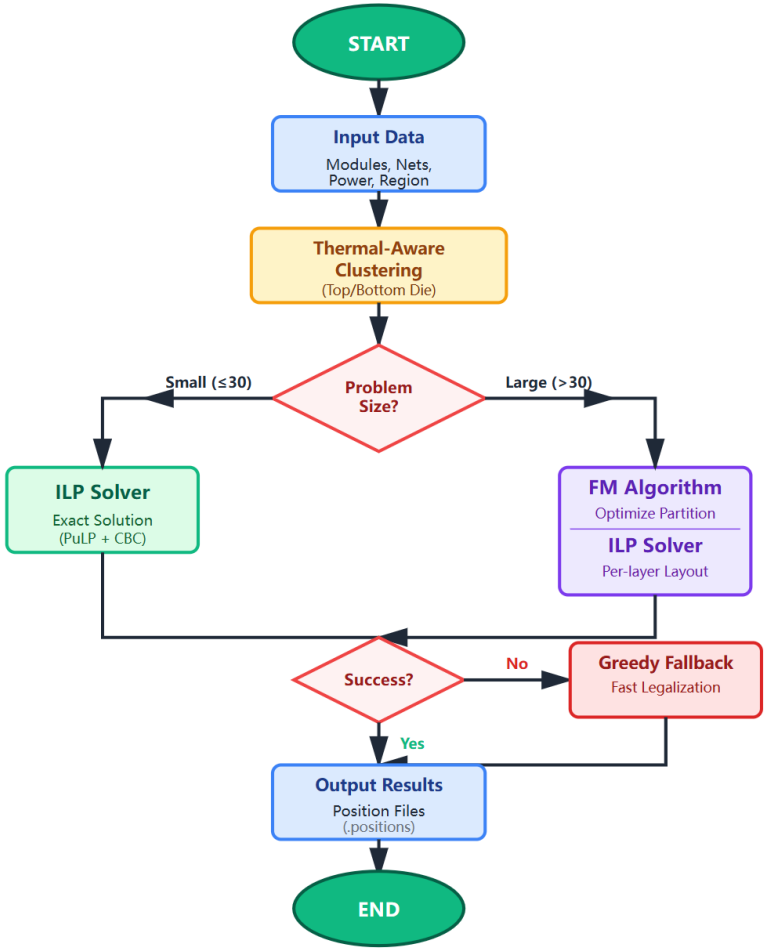
  // $\mathcal{O}_f$: fine-grained orientations (R0/R90/R180/R270)

else

  $\mathcal{O}_f \leftarrow \mathcal{O}_c$

end if

return $(\mathcal{L}, \mathcal{O}_f, \mathcal{D})$ =0



Constraints – Cross-Layer Relationship Variable

Introduce layer discontinuity variable d_{ij} (indicates *i* and *j* are on different layers):

$$d_{ij} \geq z_i - z_j$$

$$d_{ij} \geq z_j - z_i$$

$$d_{ij} \leq z_i + z_j$$

$$d_{ij} \leq 2 - z_i - z_j$$

Define same-layer indicator: $s_{ij} = 1 - d_{ij}$

When same layer, enforce minimum spacing in at least one direction:

$$\ell_{ij} + r_{ij} + b_{ij} + a_{ij} \geq s_{ij}$$

Constraints – Directional Separation

Same-layer non-overlap constraints (e.g., left side):

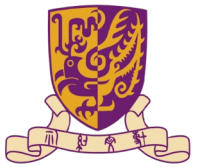
$$x_i + w^{\text{eff}}_i + \delta_{\min} \le x_j + M \cdot (1 - \ell_{ij})$$

where $\delta_{\min}$ = 0.1 mm (100 μm minimum spacing),
M is a sufficiently large constant (big-M method).

Similar constraints for other directions:

- Right: $x_j + w^{\text{eff}}_j + \delta_{\min} \le x_i + M \cdot (1 - r_{ij})$
- Bottom: $y_i + h^{\text{eff}}_i + \delta_{\min} \le y_j + M \cdot (1 - b_{ij})$
- Above: $y_j + h^{\text{eff}}_j + \delta_{\min} \le y_i + M \cdot (1 - a_{ij})$

Constraints – Thermal Penalty



For high-power modules (P_i, $P_j \geq P_{threshold}$), introduce cross-layer overlap penalty. Define auxiliary variable $o_{ij} \in \{0, 1\}$:

$$\ell^x_{ij} + r^x_{ij} + b^z_{ij} + a^z_{ij} \geq d_{ij} - o_{ij}$$

Thermal penalty:

$$\Phi_{thermal} = \sum_{\substack{i \in top, j \in bottom \\ P_i, P_j \geq P_{threshold}}} (P_i + P_j) \cdot o_{ij} + \Phi_{same\text{-}layer}$$

where $\Phi_{same\text{-}layer}$ accounts for intra-layer high-power modules that are too close (distance below threshold).

Constraints – Inter-Layer Balance

$$P_{\text{bottom}} = \sum_{i=1}^{n} P_i \cdot (1 - z_i)$$

$$P_{\text{top}} = \sum_{i=1}^{n} P_i \cdot z_i$$

$$\Delta P = |P_{\text{bottom}} - P_{\text{top}}|$$

Linearization using absolute value:

$\Delta P = p^+ + p^-$, where

$P_{\text{bottom}} - P_{\text{top}} = p^+ - p^-$

(p^+, $p^- \geq 0$ are auxiliary variables)

Algorithm Design

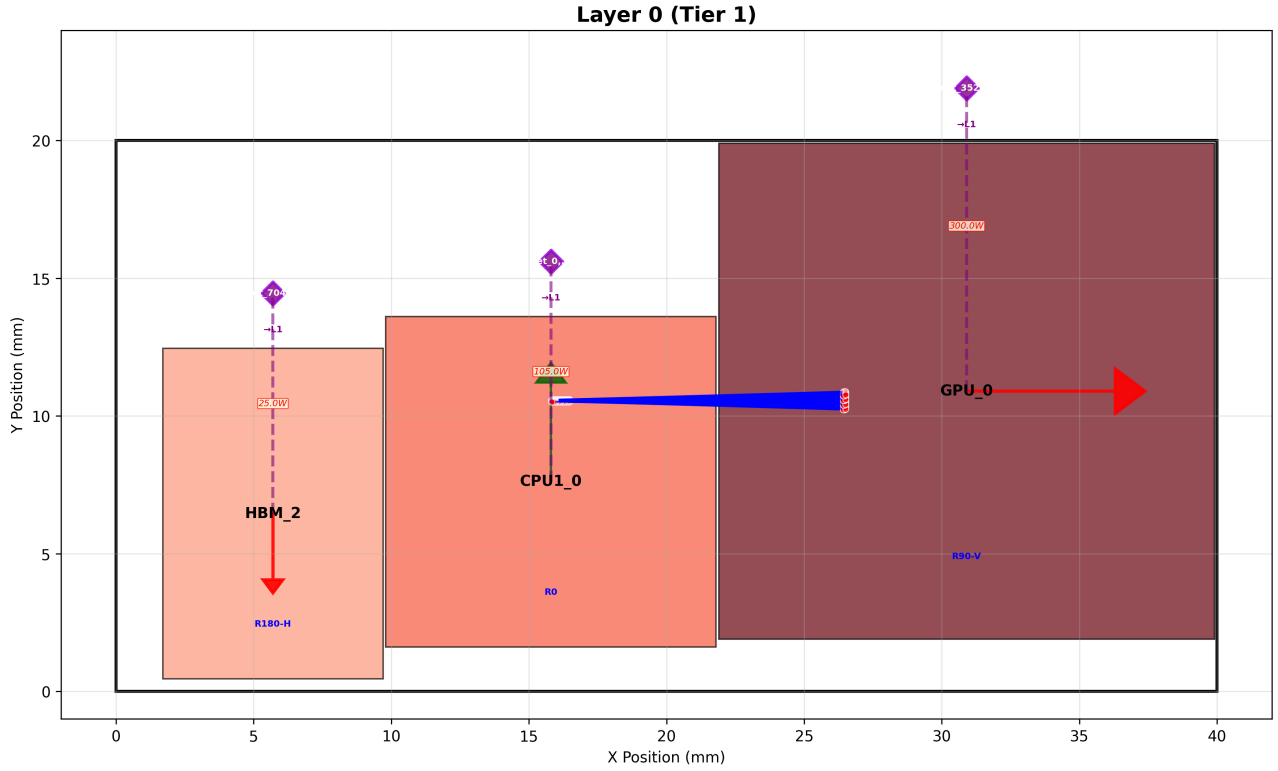
Algorithm 2 Phase 2: Pin-Aware Greedy Refinement

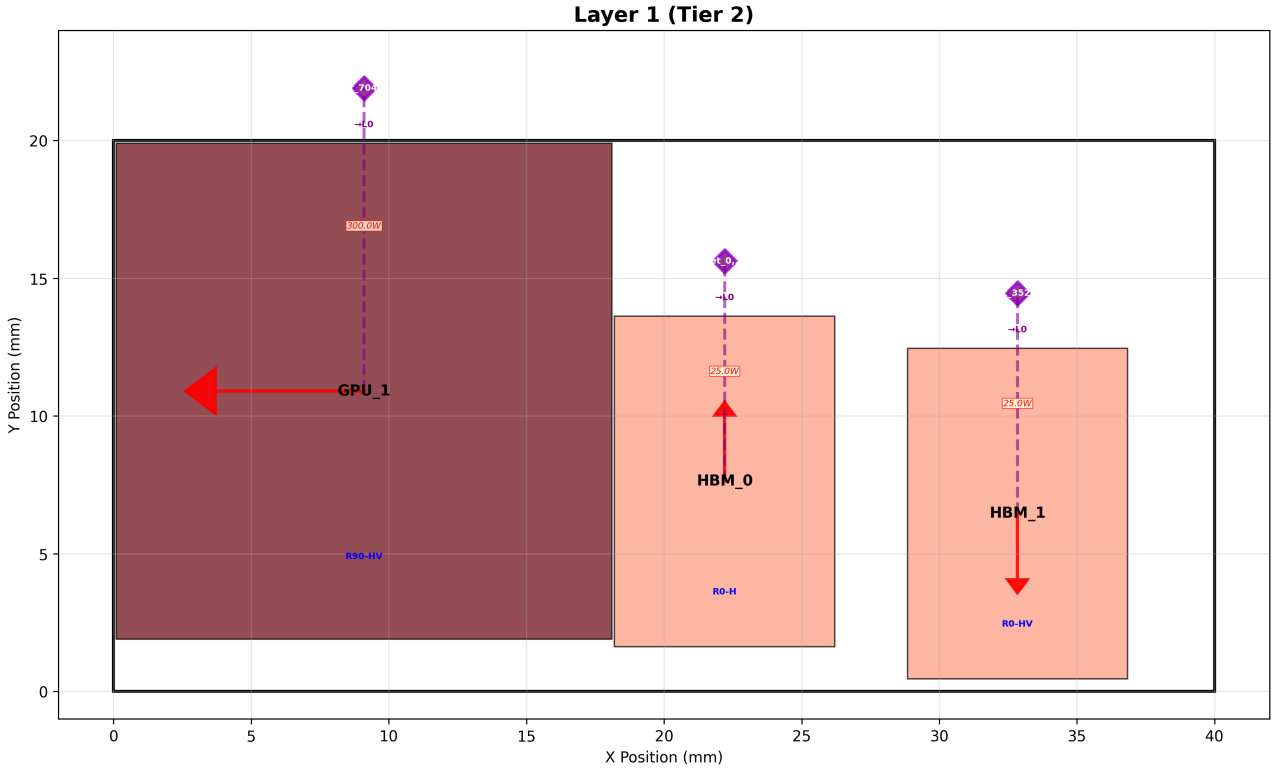
Require: Locations $\mathcal{L}$, coarse orientations $\mathcal{O}_c$, detailed nets $\mathcal{N}_d$, timeout T_2
Ensure: Fine-grained orientations $\mathcal{O}_f$

$\mathcal{O}_f \leftarrow \mathcal{O}_c$ // Initialize with Phase 1 result
$t_{\text{start}} \leftarrow$ current_time()
for each block i in $\mathcal{B}$ **do**
  if current_time() $-t_{\text{start}} > T_2$ **then**
    break // Timeout
  end if
  // Determine allowed rotations based on footprint
  if i is square **then**
    $\mathcal{R}_i \leftarrow$ {R0, R90, R180, R270}
  else if $\mathcal{O}_c[i] =$ horizontal **then**
    $\mathcal{R}_i \leftarrow$ {R0, R180} // Preserve $w \times h$ footprint
  else
    $\mathcal{R}_i \leftarrow$ {R90, R270} // Preserve $h \times w$ footprint
  end if
  // Greedy selection based on pin distance
  $r^* \leftarrow \arg\min_{r \in \mathcal{R}_i} C_{\text{pin}}(i, r)$ // Eq. (23)
  $\mathcal{O}_f[i] \leftarrow r^*$
end for
return $\mathcal{O}_f$ =0

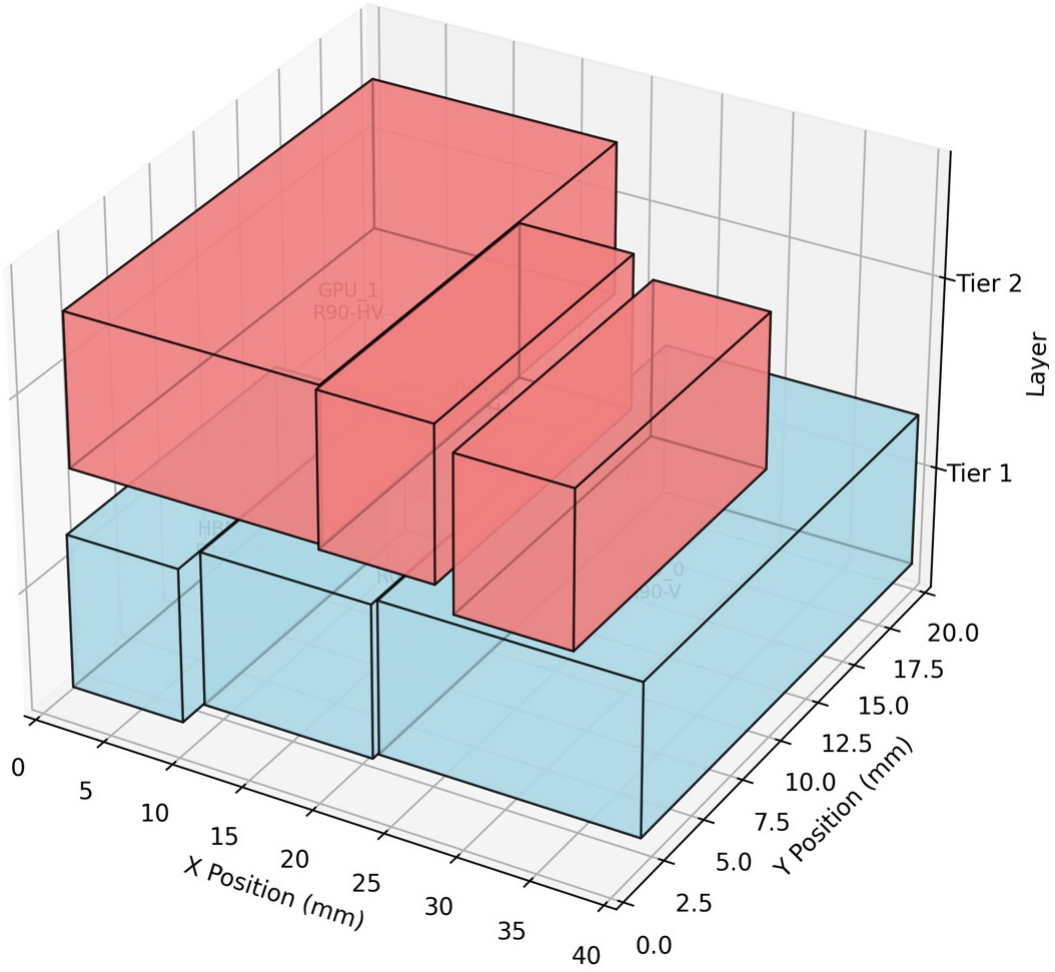
$$r_i^* = \arg\min_{r \in \mathcal{R}_i} C_{\text{pin}}(i, r) \tag{23}$$

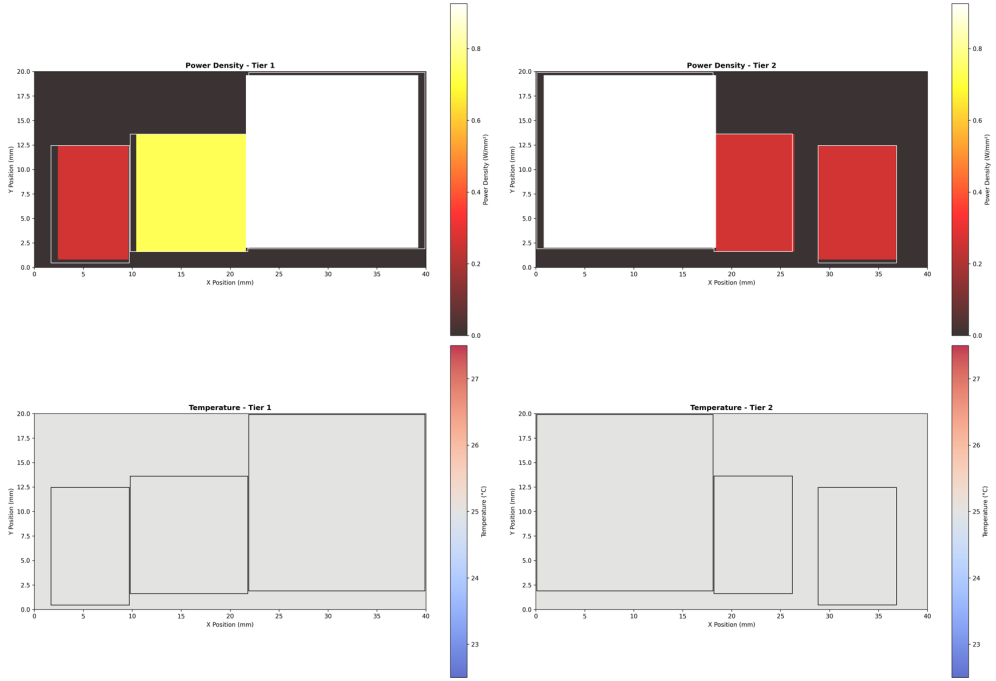
Placer Experiments – Case 1





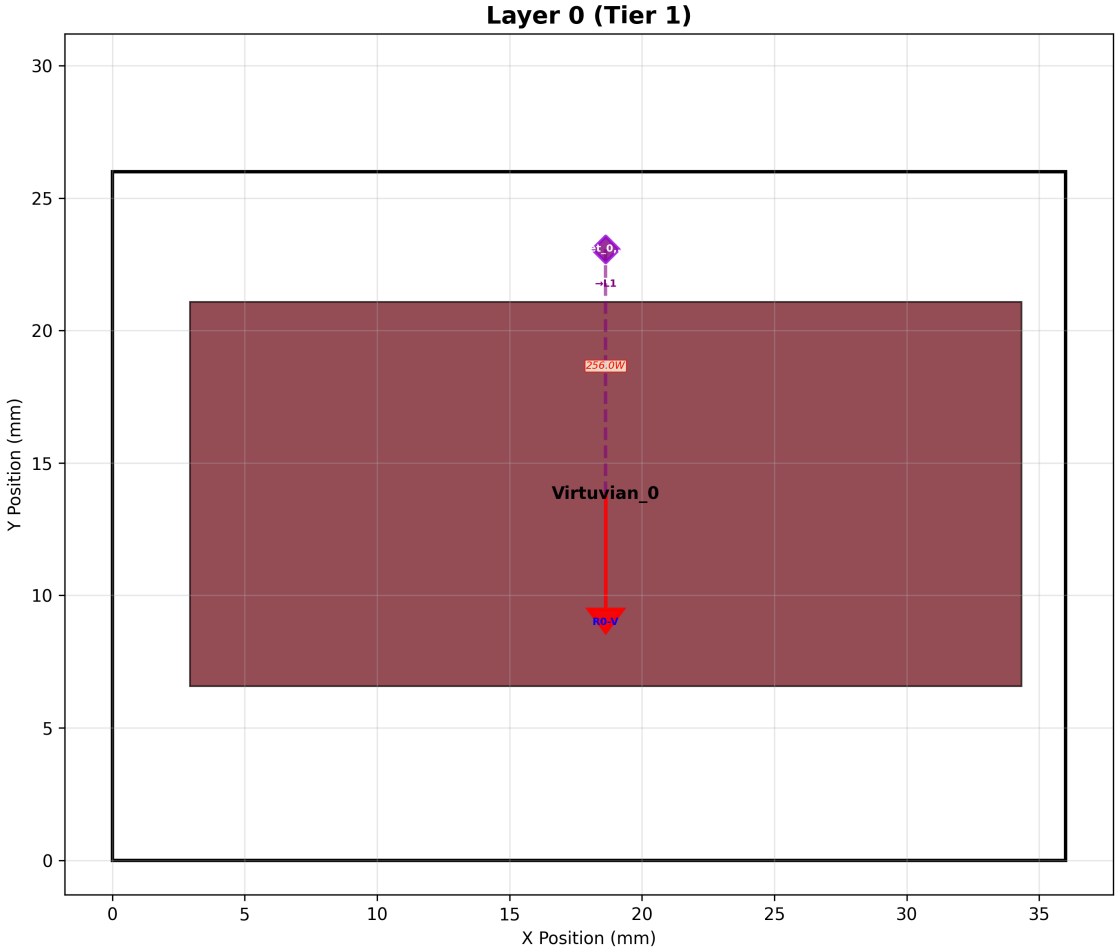
Placer Experiments – Case 1

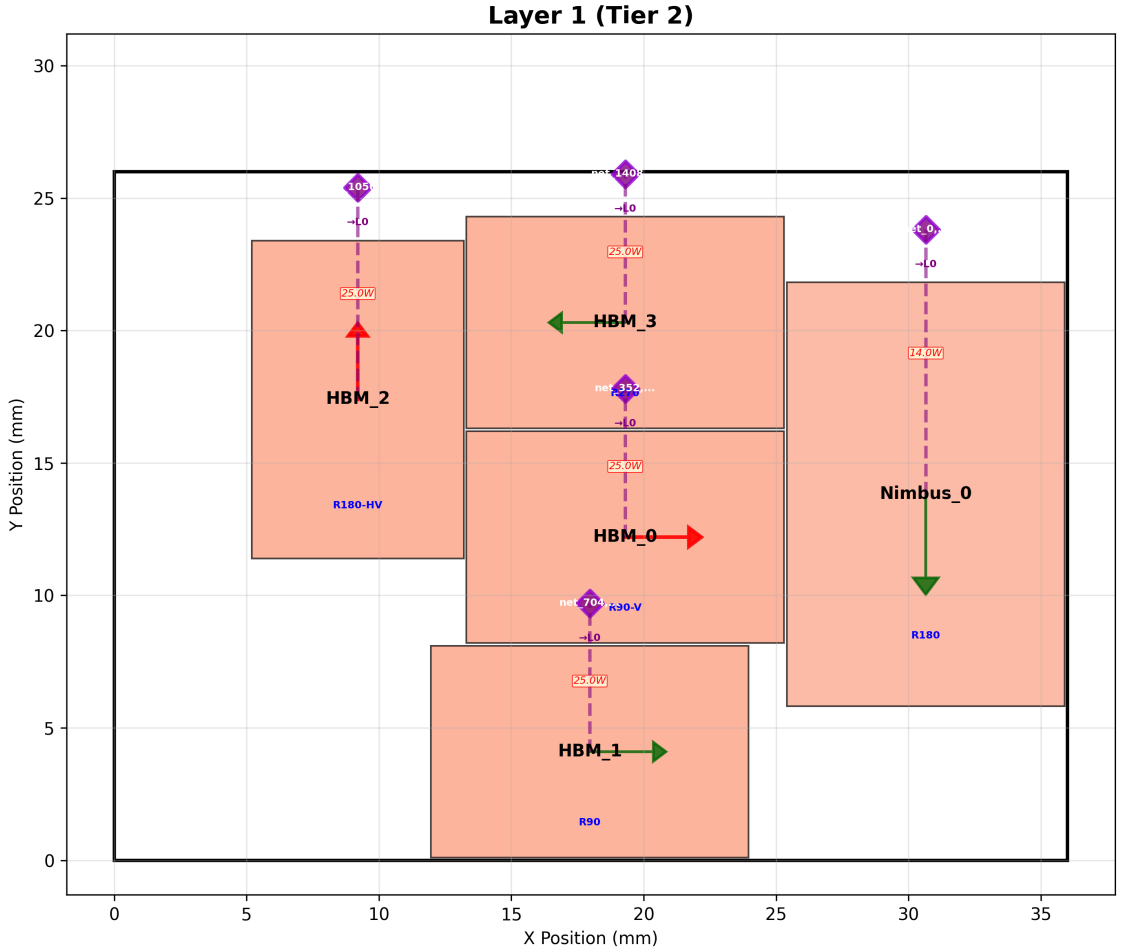




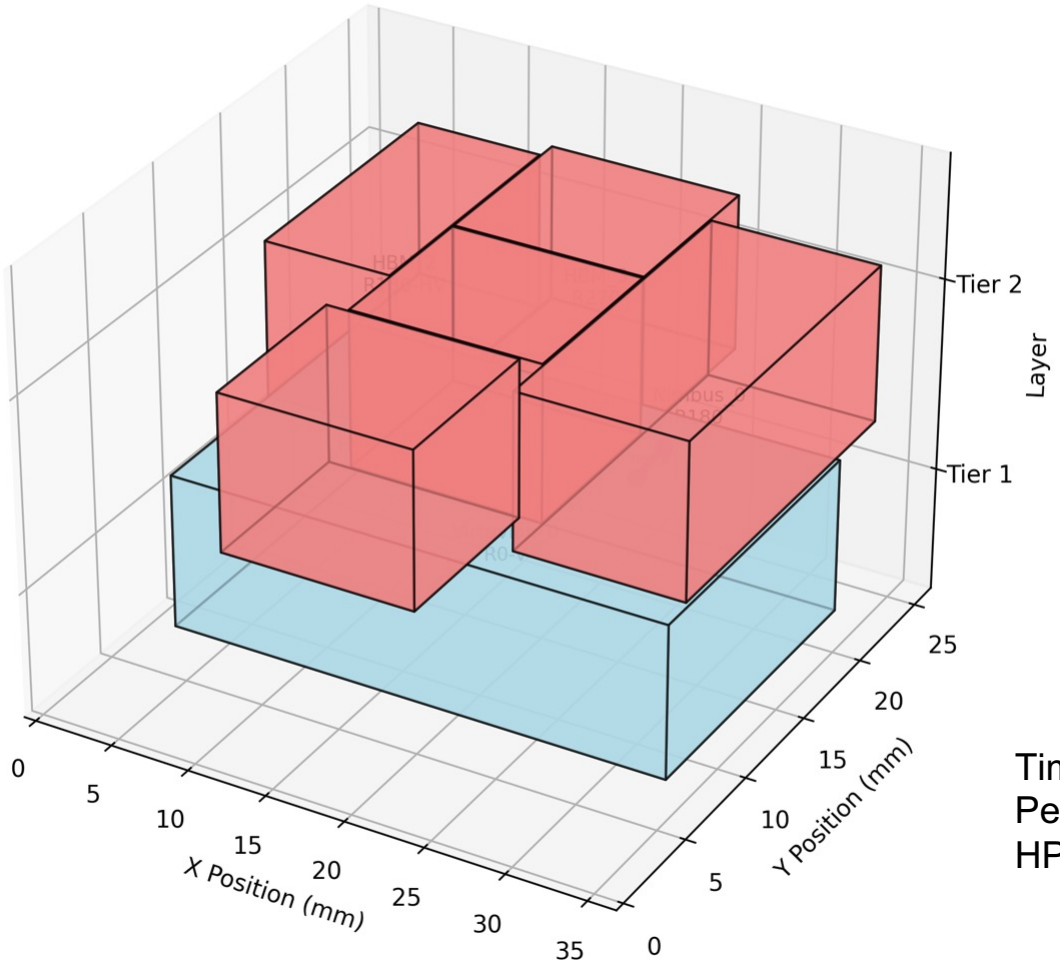
Time: 15.891757011413574
Peak Temperature: 306.103
HPWL: 5936.29 mm

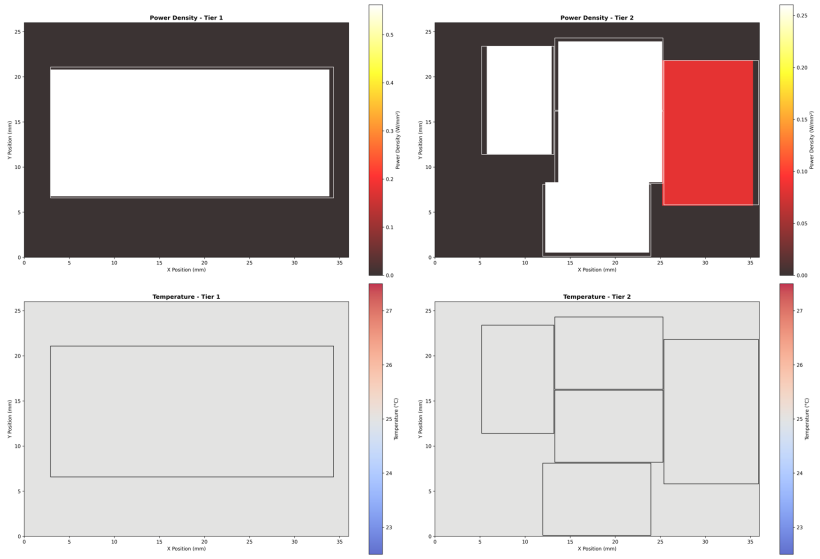
Placer Experiments – Case 2





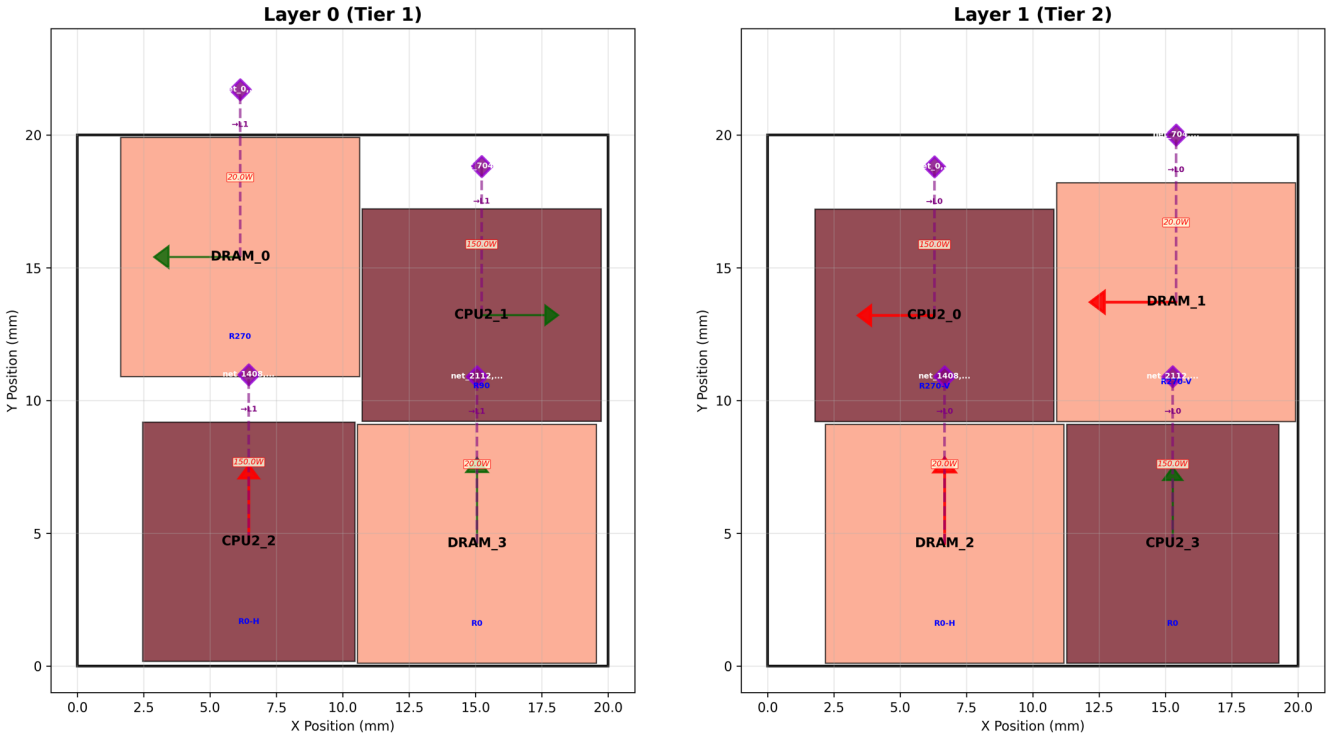
Placer Experiments – Case 2

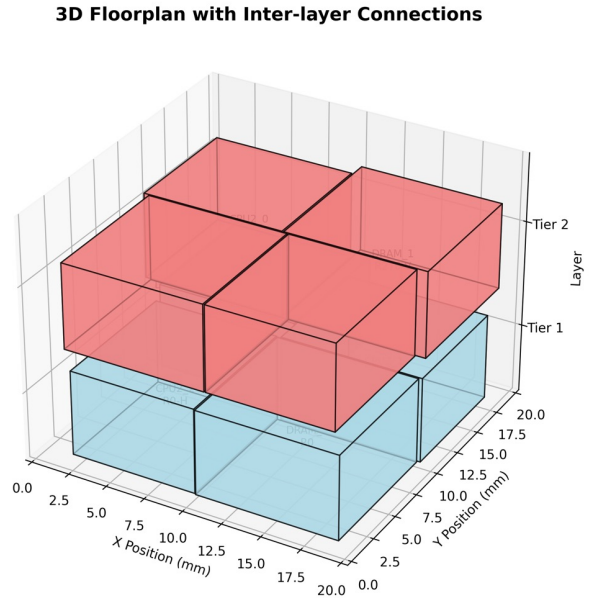


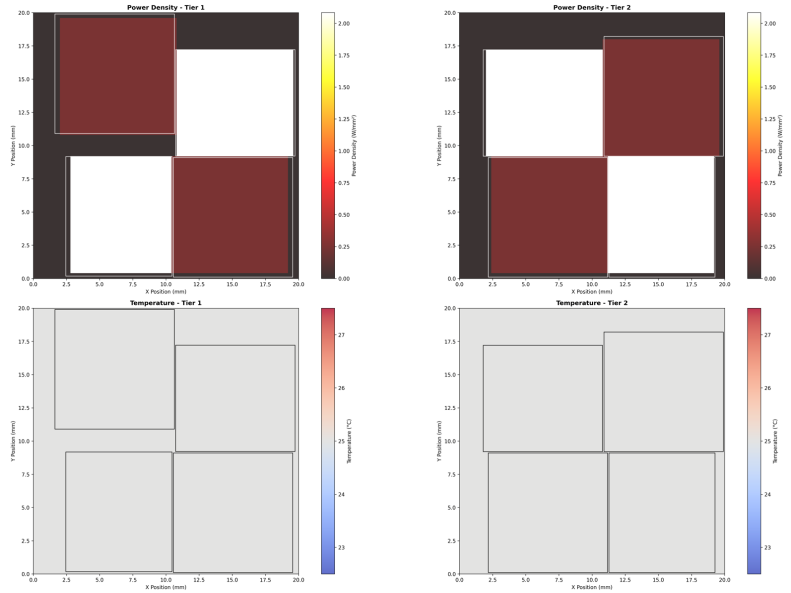


Time: 111.38
Peak Temperature: 306.05
HPWL: 5054.79 mm

Placer Experiments – Case 3

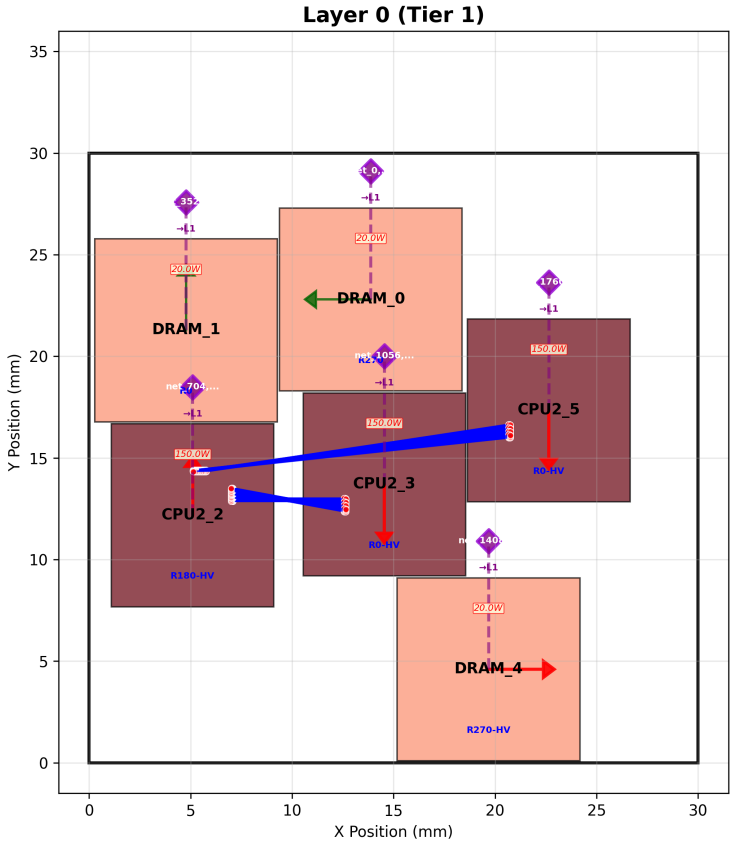


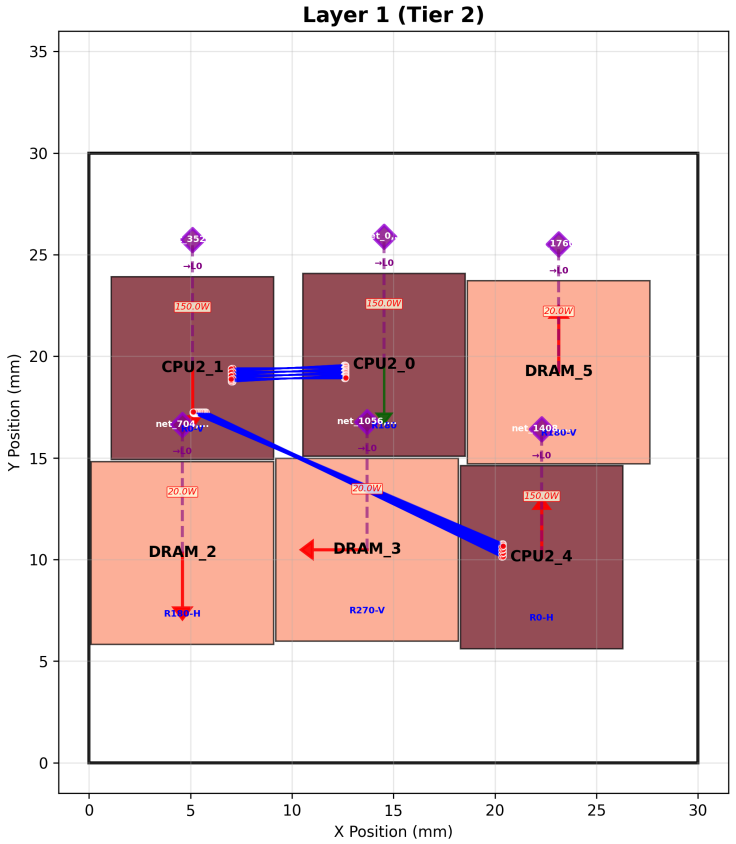


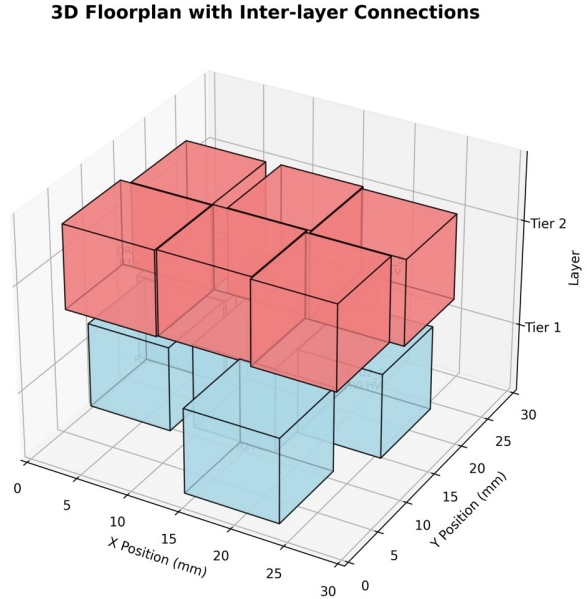


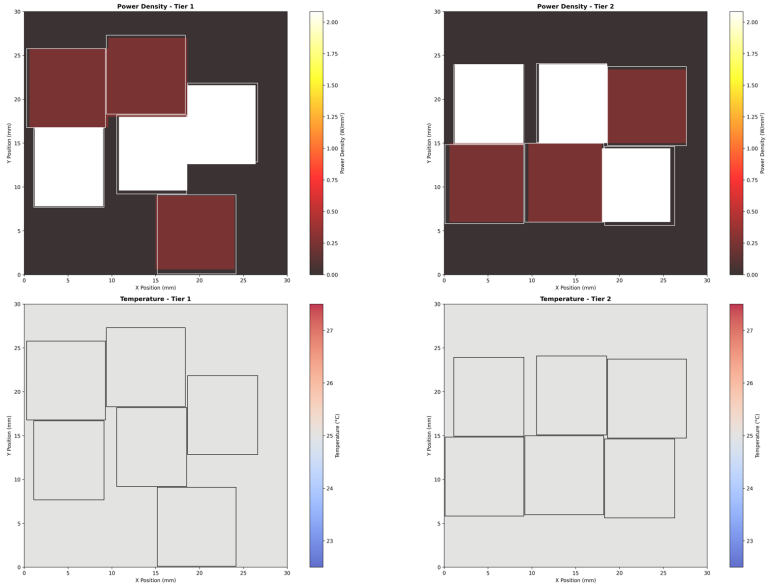
Time: 62.58
Peak Temperature: 317.46
HPWL: 10147.99 mm

Placer Experiments – Case 4



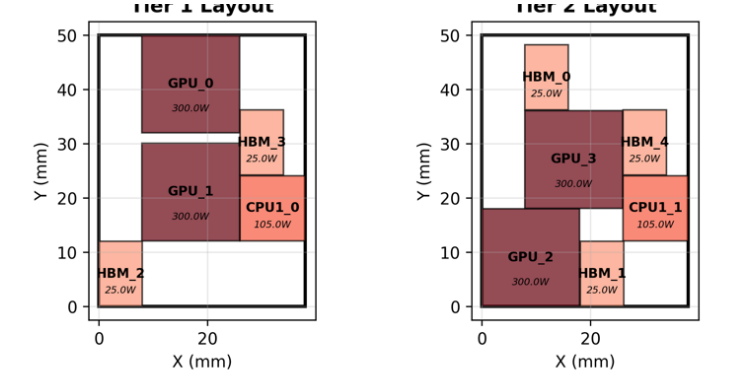


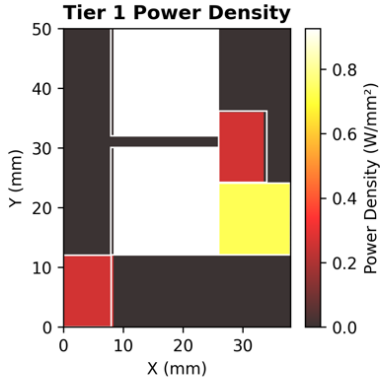


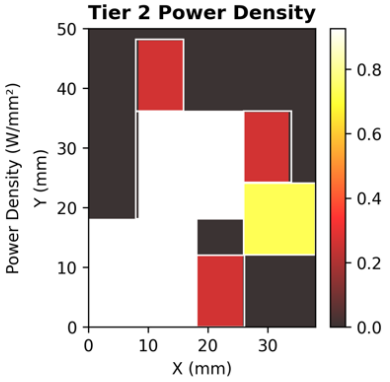


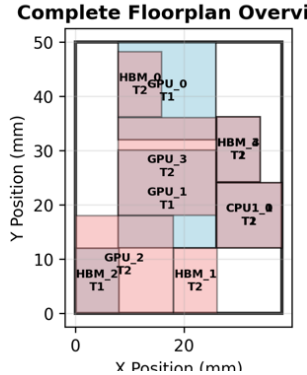
Time: 1202.22
Peak Temperature: 311.83
HPWL: 16995.43 mm

Placer Experiments – Case 5





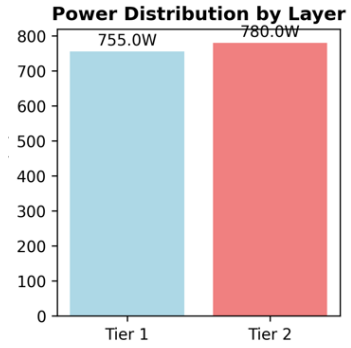


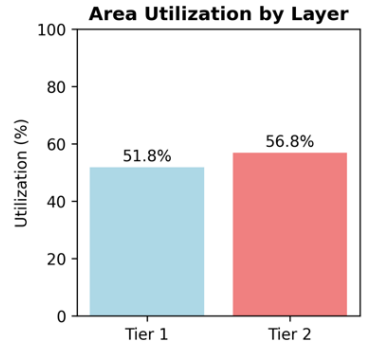


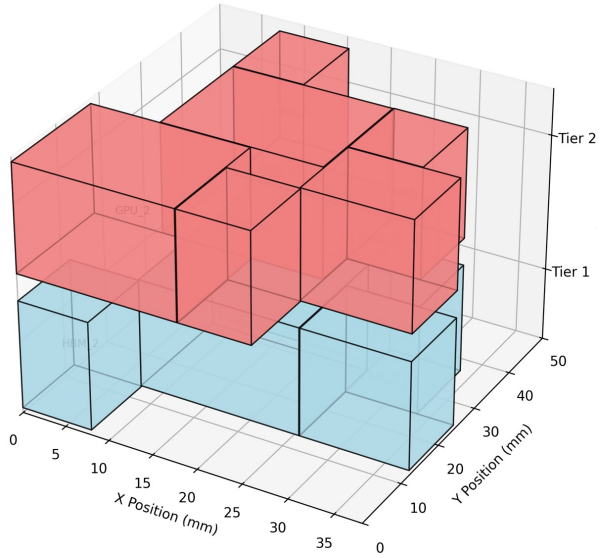

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 11
Total Nets: 3520
Layers: 2

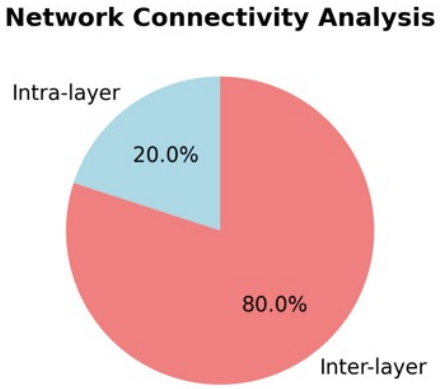
COSTS:
Total: 31221.92
Wirelength: 31171.92
Thermal: 25.00

POWER:
Total: 1535.0 W
Tier 1: 755.0 W
Tier 2: 780.0 W
```



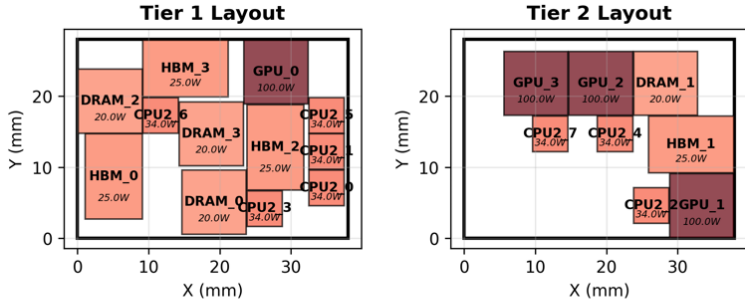


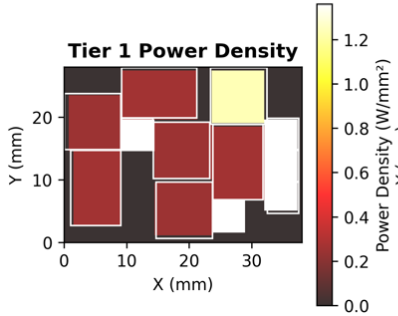


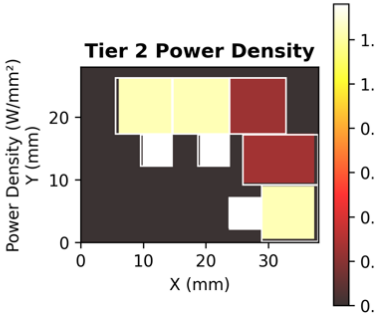


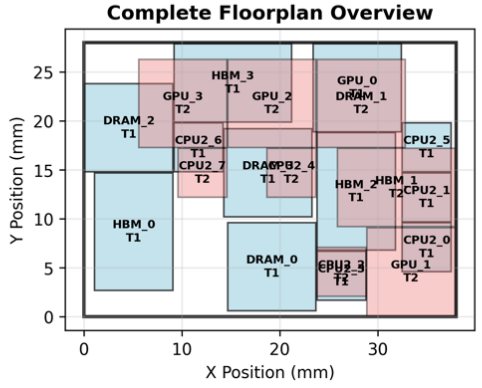
Peak Temp.315.2

Placer Experiments – Case 6





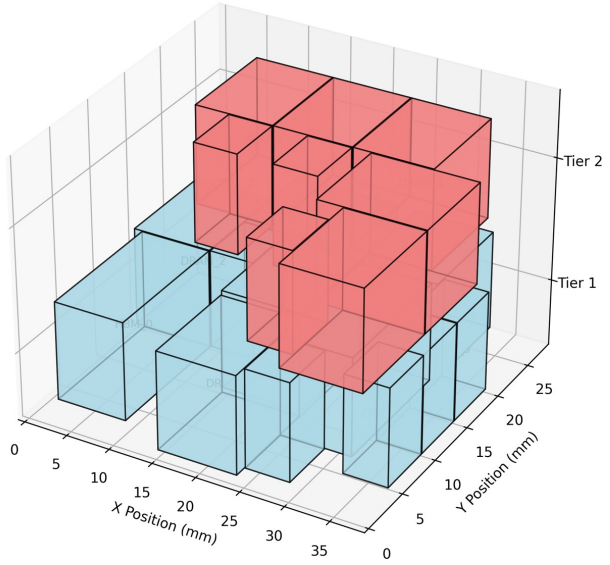


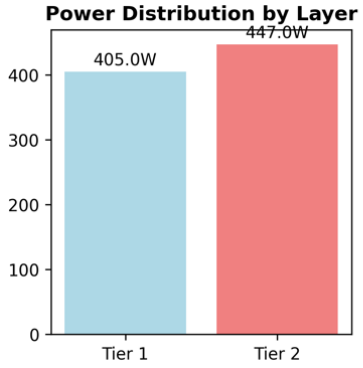


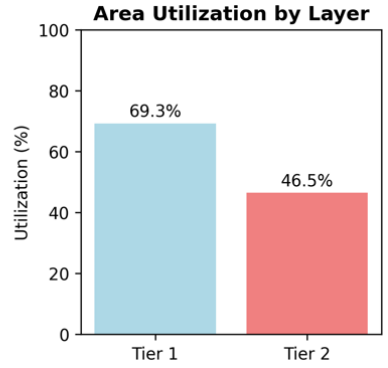

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 20
Total Nets: 2816
Layers: 2

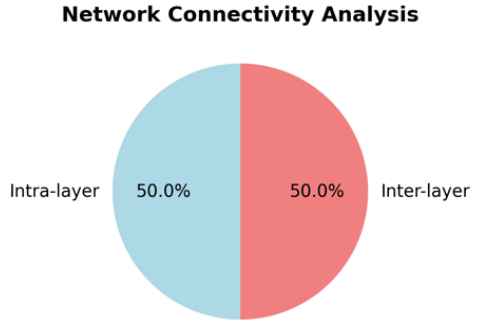
COSTS:
Total: 22488.02
Wirelength: 22438.01
Thermal: 25.00

POWER:
Total: 852.0 W
Tier 1: 405.0 W
Tier 2: 447.0 W
```



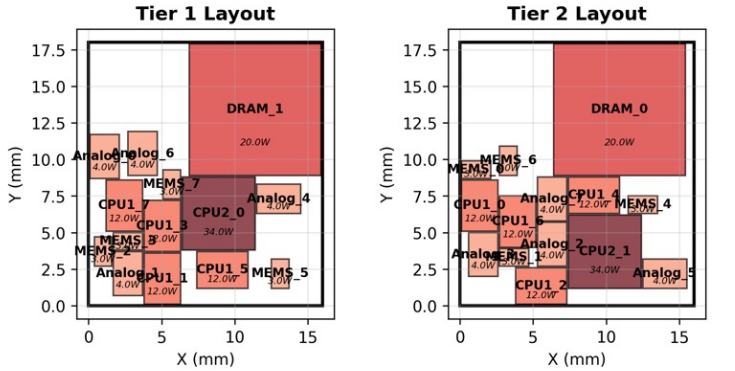


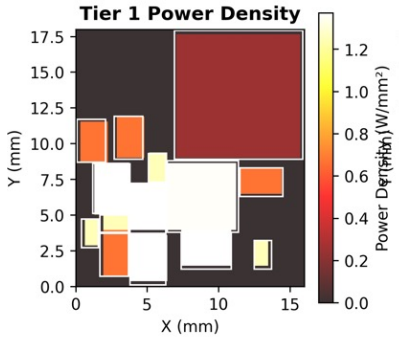


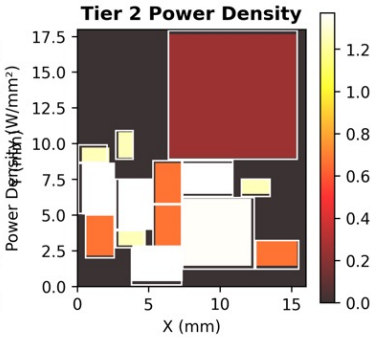


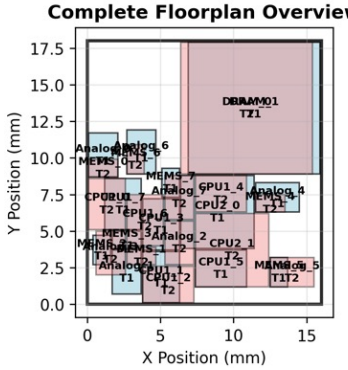
Time: 69.69
Peak Temp.: 283.39
HPWL: 22438.01 mm

Placer Experiments – Case 7





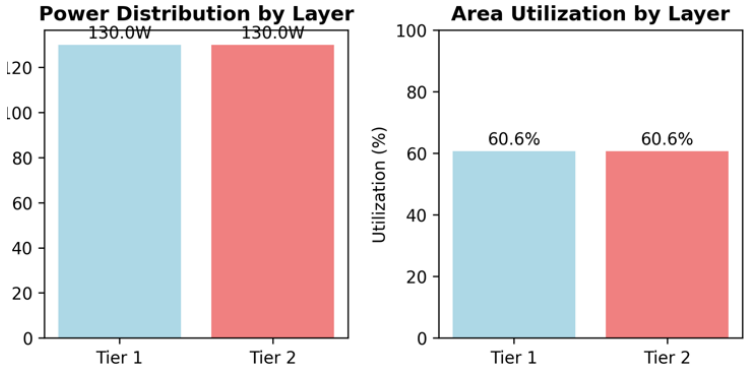


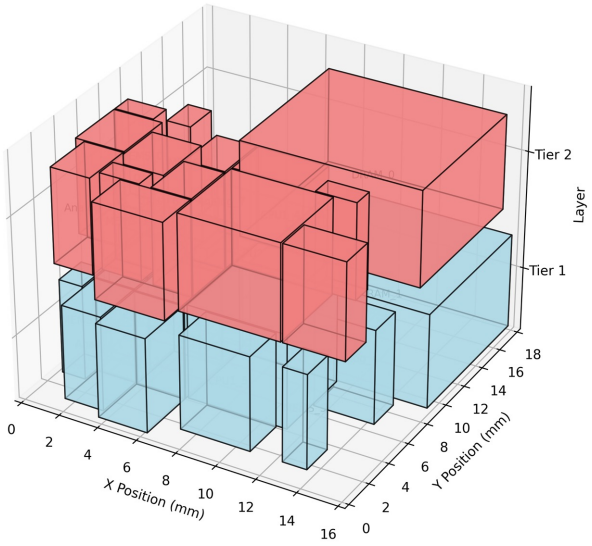


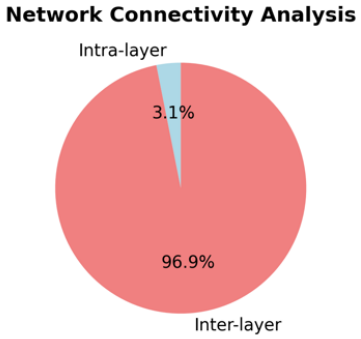

```
OPTIMIZATION STATISTICS
==========================
Total Blocks: 28
Total Nets: 1408
Layers: 2

COSTS:
Total: 4289.54
Wirelength: 4239.53
Thermal: 25.01

POWER:
Total: 260.0 W
Tier 1: 130.0 W
Tier 2: 130.0 W
```

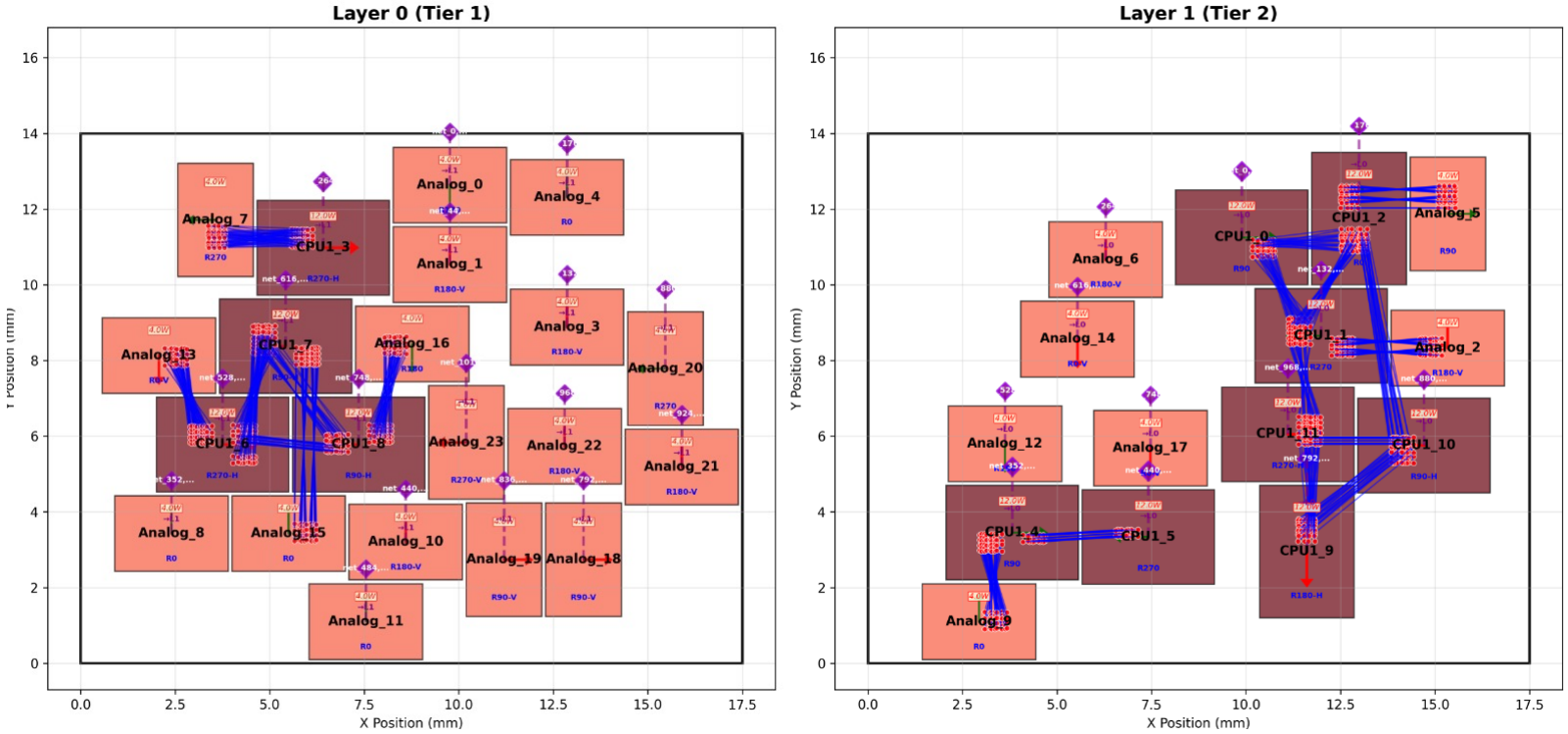


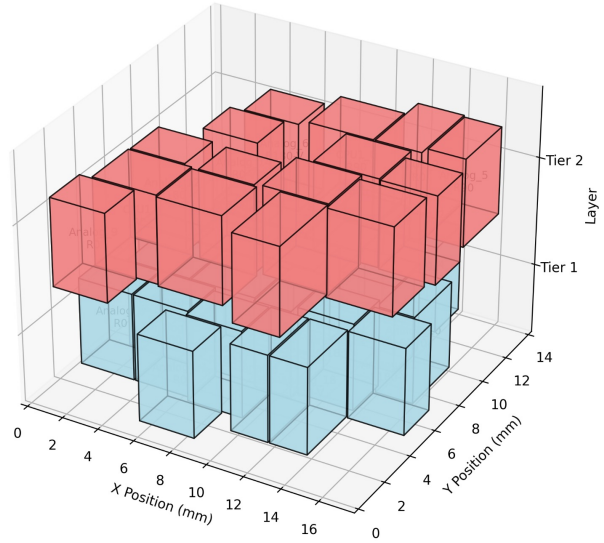




Time: 71.87
Peak Temp.: 303.32
HPWL: 4289.54 mm

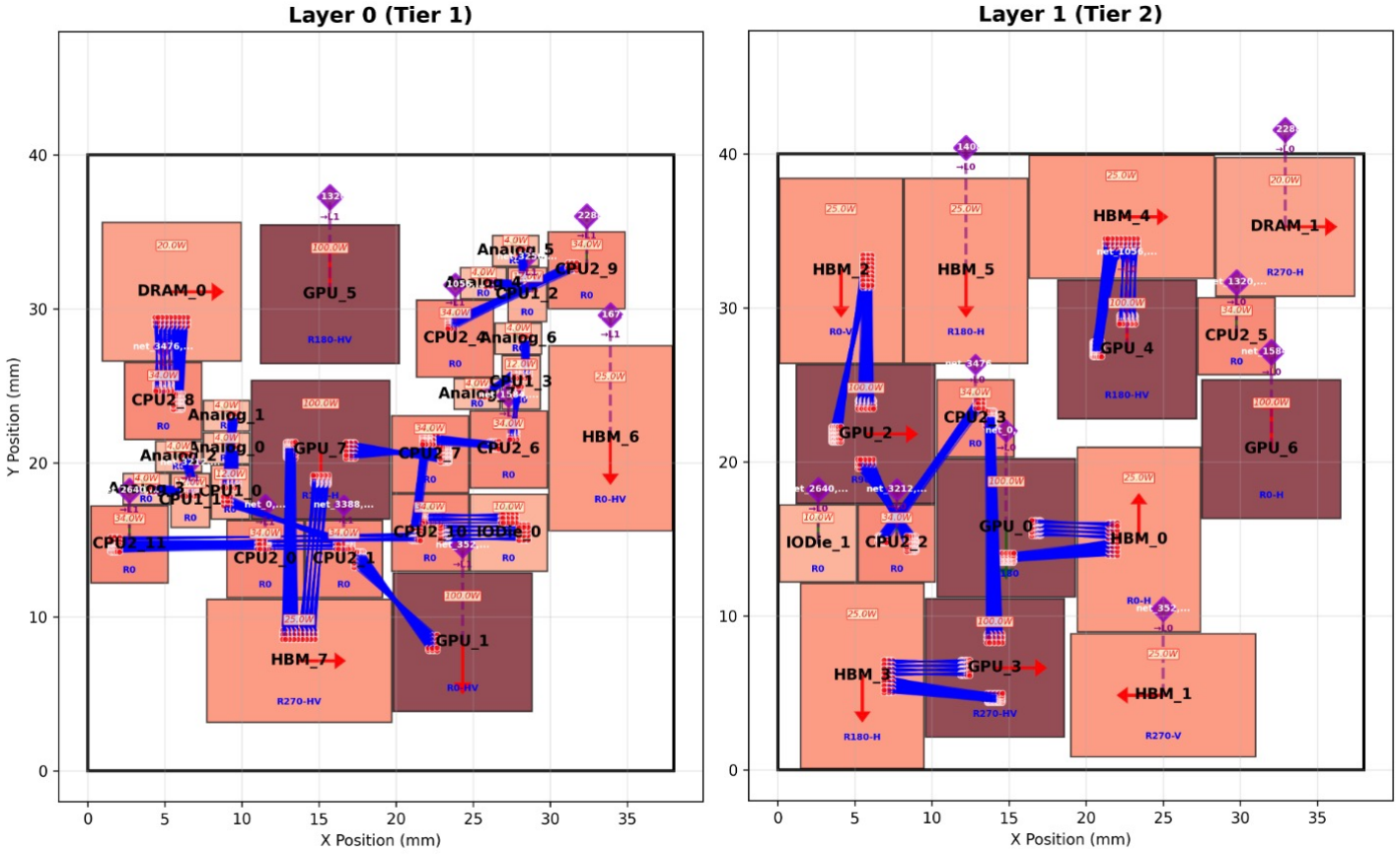
Placer Experiments – Case 8

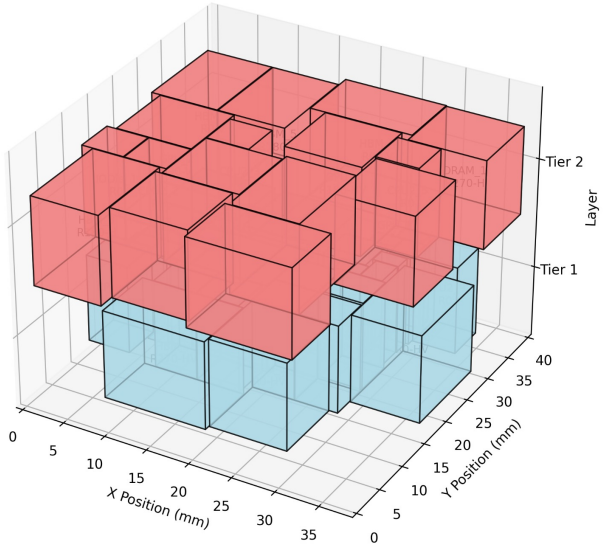




Time 3504.624584913254
Peak temp. 282.85
HPWL: 3361.93 mm

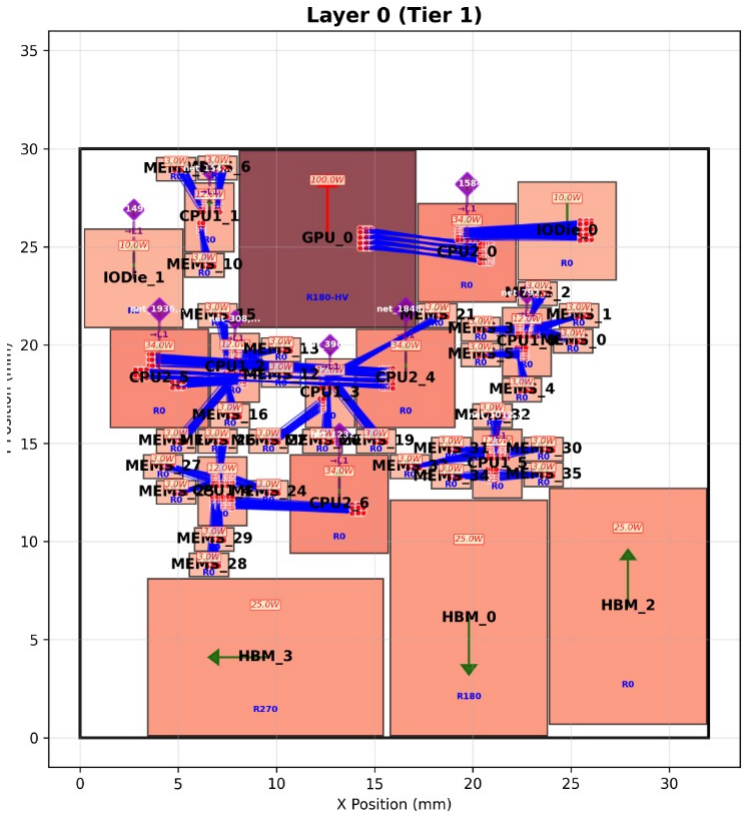
Placer Experiments – Case 9

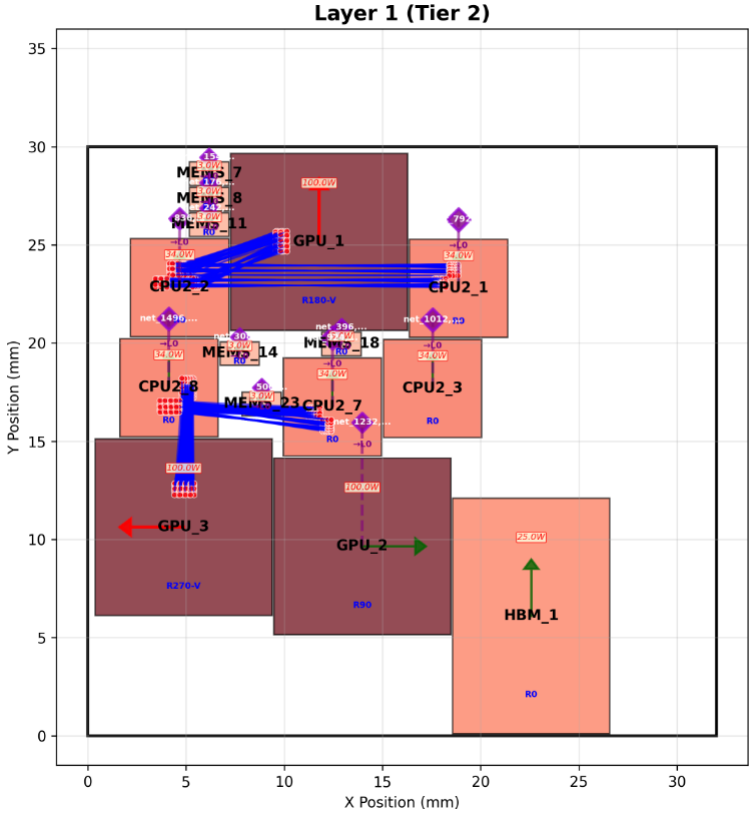


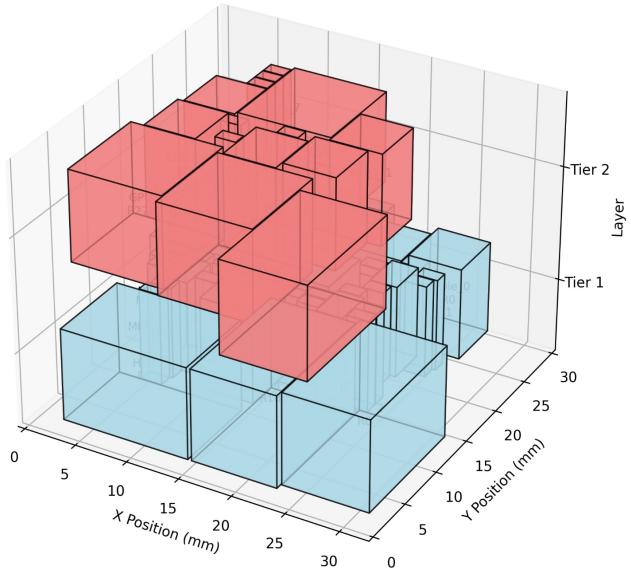


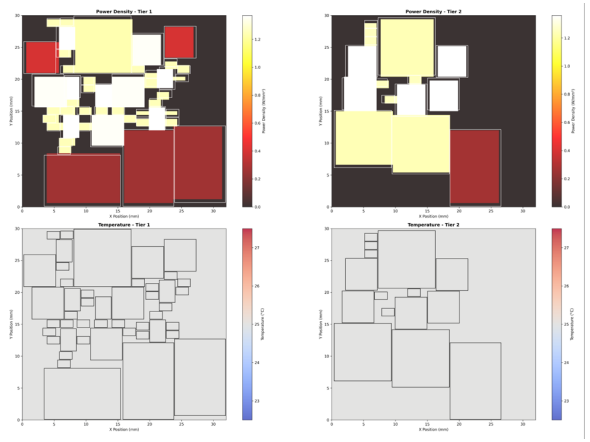
Time 3507.0527629852295
Peak temp. 334.82
HPWL: 48358.59 mm

Placer Experiments – Case 10



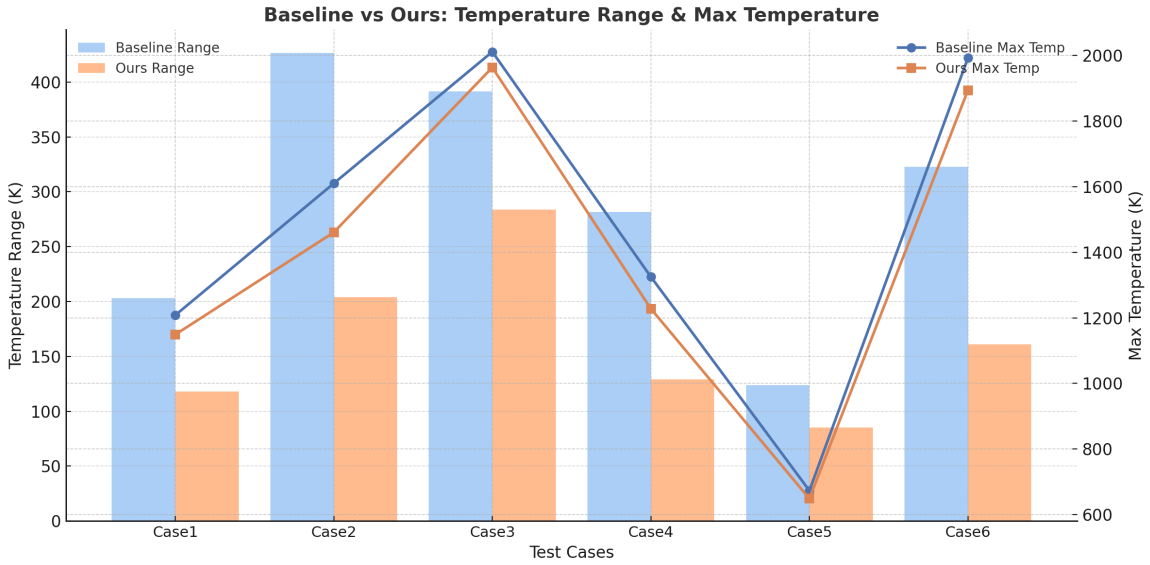






Time 3507.3127374649
Peak temp. 300.15
HPWL: 28138.81 mm

Comparison Experimental Results – Temp. Peak

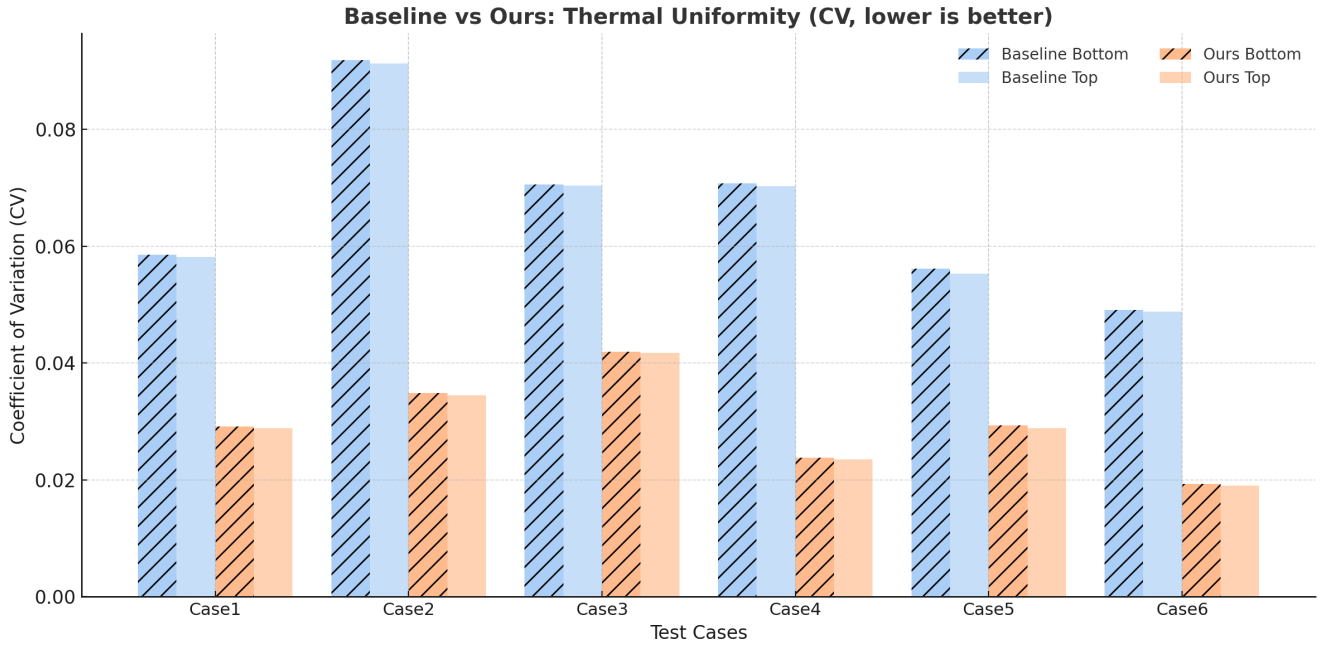


For each test case, our method get “better” results:
- Lower Max. Temperature;
- Smaller Temperature Range.

On average, peak temperature decreases by ~17K (≈1.2%), and in Case 6 our method achieves a substantial reduction of nearly 100K compared with the baseline.

Ours	Bottom Die (K)			Top Die (K)			Baseline	Bottom Die (K)			Top Die (K)		
	Min.	Max.	Range	Min.	Max.	Range		Min.	Max.	Range	Min.	Max.	Range
1	1030.89	1148.89	118	1030.86	1147.73	116.87	1	1030.89	1148.89	118	1030.86	1147.73	202.06
2	1256.43	1460.58	204.15	1256.36	1457.89	201.53	2	1256.43	1460.58	204.15	1256.36	1457.89	423.94
3	1679.34	1963.14	283.8	1679.31	1961.91	282.6	3	1679.34	1963.14	283.8	1679.31	1961.91	390.13
4	1098.67	1227.89	129.22	1098.65	1226.17	127.52	4	1098.67	1227.89	129.22	1098.65	1226.17	280.05
5	563.78	648.94	85.16	563.76	647.44	83.68	5	563.78	648.94	85.16	563.76	647.44	122.23
6	1732.74	1893.58	160.84	1732.71	1890.03	157.32	6	1669.33	1991.97	322.64	1669.31	1990.27	320.96

Experimental Results – Temp. Balanced Distribution

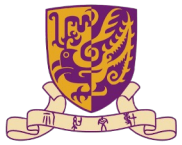


We use Coefficient of Variation (CV) to evaluate the balanced or not of the die temperature distribution.

$$CV = \frac{\sigma}{\mu}$$

Compared with the baseline, our method improves thermal uniformity by reducing the coefficient of variation (CV) by approximately 55.1%.

	Ours				Baseline			
	Bottom Die		Top Die		Bottom Die		Top Die	
Test case	Mean.	Std.	Mean.	Std.	Mean.	Std.	Mean.	Std.
1	1079.874	31.49698	1079.466	31.16869	1080.114	63.22427	1079.611	62.75536
2	1310.281	45.66117	1309.928	45.18138	1310.527	120.3783	1309.996	119.519
3	1788.517	75.05186	1788.027	74.68629	1788.06	126.1314	1787.677	125.717
4	1143.345	27.165	1142.97	26.86789	1143.296	80.84383	1142.92	80.35285
5	587.1468	17.23045	586.723	16.92625	587.2032	32.95235	586.7746	32.44569
6	1806.552	34.86248	1805.685	34.31764	1806.163	88.57475	1805.684	88.1785



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Thanks for Your Attention!